**L.J Institute of Engineering and Technology**

**Data Structures using JAVA PRACTICE BOOK (SEM-II EVEN_2025-CE RELATED BRANCHES)**

**Note :**

**This practice book is only for reference purpose . L.J.U Test question paper may not be completely set from this practice book.**

**Date : 28 May 2025**

| Sr No | Unit Number | Question_Text | MCQ Answer | Marks | Option A | Option B | Option C | Option D |
|---|---|---|---|---|---|---|---|---|
| 1 | 7 | In doubly linked lists, traversal can be performed? | C | 1 | Only in forward direction | Only in reverse direction | In both directions | None |
| 2 | 7 | Which amongst the following segment of code counts the number of elements in the non empty doubly linked list, if it is assumed that X points to the first element of the list and ctr is the variable which counts the number of elements in the list? | C | 1 | for(ctr=1; X!=null; ctr++) X=X.fwd | for(ctr=1; X!=null; ctr++) X=X.bwd | for(ctr=1; X.fwd!=null; ctr++) X=X.fwd | for(ctr=1; X.bwd!=null; ctr++) X=X.bwd |
| 3 | 7 | In a doubly linked list the number of pointers affected for an insertion operation (insert before particular value) will be …. (Assume that the value present in a doubly LinkedList but not on the head node) | A | 1 | 4 | 0 | 1 | depends upon the nodes of doubly linked list |
| 4 | 7 | Which type of linked list stores the address of the head node in the next pointer of the last node? | C | 1 | singly linked list | doubly linked list | circular linked list | stack implemented using linked list |
| 5 | 7 | Which of the following is false about a doubly linked list? | D | 1 | We can navigate in both the directions | It requires more space than a singly linked list | The insertion and deletion of a node take a bit longer | Implementing a doubly linked list is easier than singly linked list |
| 6 | 7 | Which of the following operations is performed more efficiently by doubly linked list than by linked list singly? | A | 1 | Deleting a node whose location in given | Searching of an unsorted list for a given item | Inverting a node after the node with given location | Traversing a list to process each node |
| 7 | 7 | A node in a double linked list comprises of: | C | 1 | information field | information field and next pointer | information field, pointer to the next node in sequence, pointer to the previous node | information field, next pointer, previous pointer & previous pointer and thread field |
| 8 | 7 | In circular linked list(implemented using singly linked list), nodes contains _____fields and In Doubly linked list, nodes contains ____fields? | D | 1 | 2, 2 | 3, 2 | 1, 2 | 2, 3 |
| 9 | 7 | A linear list in which each node has pointer to the previous and next node is called? | D | 1 | singly Linked list | singly circular Linked list | circular Linked list | Doubly Linked list |
| 10 | 7 | If a Doubly linked list is declared as<br>class Node<br>{ int info;<br>Node Fwd;<br>Node Bwd;<br>}<br>Then to insert a node in the middle of the list, requires how many changes to various next and prev pointers? | A | 1 | 2 next, 2 prev | 1 next, 2 prev | 1 next, 1 prev | 2 next, 1 prev |

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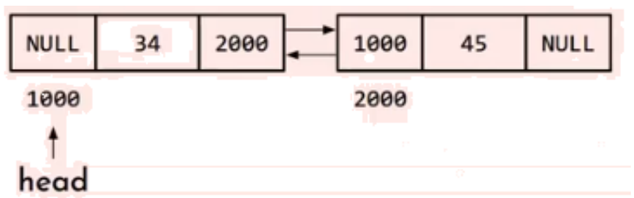
|---|---|---|---|---|---|---|---|---|
| 11 | 7 | What does the following code snippet do(Consider the required class is already created)?<br>Node solve(Node head) {<br>Node prev = null;<br>if(head == null) {<br>return head; }<br>if(head.next == null) {<br>return head;}<br>Node curr = head.next;<br>while(head != null) {<br>head.next = prev;<br>prev = head;<br>head = curr;<br>if(curr != null) {<br>curr = curr.next;<br>}}<br>return prev; } | B | 1 | Returns the original linked list | Returns the linked list after reversing it | Returns a linked list containing elements at odd indices only | print all the nodes of linked list |
| 12 | 7 | How do you count number of nodes in circular linked list? | B | 1 | int count (Node head)<br>{<br>Node t=head;<br>int c=0;<br>if ( t = = null)<br>return 0;<br>while (t != NULL)<br>{<br>c++;<br>t = t .next;<br>}<br>return c;} | int count (Node head)<br>{<br>Node t=head.next;<br>int c=1;<br>if ( t= = null)<br>return 0;<br>while (t != head)<br>{<br>c++;<br>t = t.next;<br>}<br>return c;} | int count (Node head)<br>{<br>Node t=head;<br>int c=0;<br>if ( t= = null)<br>return 0;<br>while (t != head.next)<br>{<br>c++;<br>t = t.data;<br>}<br>return c;} | int count (Node head)<br>{<br>Node head=t;<br>int c=0;<br>if ( t= = null)<br>return 0;<br>while (t != NULL)<br>{<br>c++;<br>t = t.next;<br>}<br>return c;} |
| 13 | 7 | Consider the following Doubly Linked List:<br><br>If head points to the first node of the linked list then what will be the output of the following code snippet?<br><br>head=head.next.next.next.prev;<br>head.next.next.prev=head;<br>S.O.P(head.next.next.prev.next.value);<br>1 --> 3 --> 5 --> 7 --> 9 --> NULL | D | 1 | 1 | 3 | 5 | 7 |
| 14 | 7 | Which of the following segment of code adds a new node in front of the last node of doubly linked list? (Assume that new is pointing to the new node to be added)<br>  <br> | B | 1 | temp=head; temp=temp.next; new.next=temp; temp.new=head; | temp= head;<br>new.prev=temp;<br>new.next=temp.next;<br>temp.next.prev=new;<br>temp.next=new; | temp= head; head.next=new; | New.next=temp.next.prev; |

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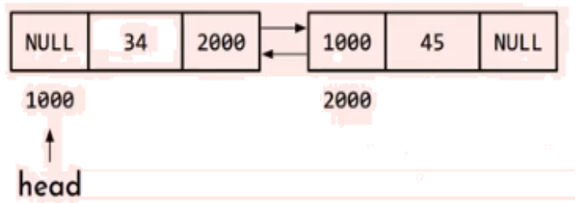
|---|---|---|---|---|---|---|---|---|
| 15 | 7 | Consider the following doubly linked list:<br>How the resultant linked list looks like after performing the following sequence of operation?<br>  <br><br>temp=head; temp= temp.next; temp.data= temp.prev.data; | C | 1 | 34->45 | 45->34 | 34->34 | 45->45 |
| 16 | 7 | A doubly linked list is declared as<br>class Node {<br>int Value;<br>Node Fwd;<br>Node Bwd;<br>};<br>Where Fwd and Bwd represent forward and backward link to the adjacent elements of the list. Which of the following segments of code deletes the node pointed by X from the doubly linked list, if it is assumed that X points to neither the first nor the last node of the list? | A | 1 | X.Bwd.Fwd = X.Fwd; X.Fwd.Bwd = X.Bwd ; | X.Bwd.Fwd = X.Fwd ;<br>X.Fwd.Fwd = X.Bwd ; | X.Bwd.Fwd = X.Bwd ;<br>X.Fwd.Bwd = X.Bwd ; | X.Bwd.Fwd = X.Bwd ;<br>X.Fwd.Bwd = X.Fwd; |
| 17 | 7 | Consider the following function that takes reference to head of a Doubly Linked List as parameter. Assume that a node of doubly linked list has previous pointer as prev and next pointer as next.<br>void fun(Node head_ref){<br>Node temp = null;<br>Node current = head_ref;<br>while (current != null){<br>temp = current.prev;<br>current.prev = current.next;<br>current.next = temp;<br>current = current.prev; }<br>if(temp != null )<br>head_ref = temp.prev;}<br>Assume that reference of head of following doubly linked list is passed to above function 1 <--> 2 <--> 3 <--> 4 <--> 5 <-->6. What should be the modified linked list after the function call? | C | 1 | 2 <--> 1 <--> 4 <--> 3 <--> 6 <-->5 | 5 <--> 4 <--> 3 <--> 2 <--> 1 <-->6 | 6 <--> 5 <--> 4 <--> 3 <--> 2 <--> 1 | 6 <--> 5 <--> 4 <--> 3 <--> 1 <--> 2 |

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|---|---|---|---|---|---|---|---|---|
| 18 | 7 | Consider the following function that takes reference to head of a Doubly Linked List as parameter named node then check the following code.<br>Node reverse(Node node){<br>Node prev = null;<br>Node current = node;<br>Node next = null;<br>while (current != null) {<br>next = current.next;<br>current.next = prev;<br>prev = current;<br>current = next; }<br>/*ADD A STATEMENT HERE*/<br>return node; }<br>What should be added in place of "/*ADD A STATEMENT HERE*/", so that the function correctly reverses a linked list. | C | 0.5 | head = prev; | current = prev; | node = prev; | head = null; |
| 19 | 7 | What the following function will do ?<br>void function(int data){<br>Node temp = new Node(data);<br>if(head == null)<br>{<br>head = temp;<br>head.next = head;<br>}<br>else<br>{<br>Node cur = head;<br>while(cur.next != head)<br>cur = cur.next;<br>temp.next = head;<br>head = temp;<br>cur.next = temp;<br>} } | C | 1 | insert a node at last | delete a node from last | insert a node at Head | remove head |
| 20 | 7 | Write a Java function that delete odd nodes from the doubly linked list. | | 2 | | | | |
| 21 | 7 | Write a method to find midpoint from Doubly Link list. | | 3 | | | | |
| 22 | 7 | Write JAVA functions to implement DELETE_FIRST_NODE and TRAVERSE operations in doubly linked list. | | 5 | | | | |
| 23 | 7 | Write an function to delete particular element from a doubly link list | | 5 | | | | |
| 24 | 7 | Write a program to insert a new node into orderly doubly linked list. | | 5 | | | | |
| 25 | 7 | Write a program to insert a node at last position in doubly linked list. | | 5 | | | | |
| 26 | 7 | Write user defined function to insert and delete a node at/from end in doubly linked list. | | 5 | | | | |
| 27 | 7 | Write a java function to calculate the sum of all data elements of all nodes from given circular linked list. | | 1 | | | | |
| 28 | 7 | Write a JAVA function to find maximum element from doubly linked list. | | 5 | | | | |
| 29 | 7 | Write functions to: (1) insert a node at the end (2) delete a node from the beginning of a doubly linked list. | | 5 | | | | |
| 30 | 7 | Write program to perform INSERT_FIRST (to insert a node at the first position) and REVERSE_TRAVERSE (to display the data in nodes in reverse order) operations in doubly linked list. | | 5 | | | | |
| 31 | 7 | Write a Java method to print only those node values which are odd, from a doubly LinkedList. | | 2 | | | | |
| 32 | 7 | Is it possible to check whether the given linked list is either NULL-terminated or ends in a cycle (cyclic)? Describe any method if exists and justify your answer. | | 1 | | | | |

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|---|---|---|---|---|---|---|---|---|
| 33 | 7 | Write a JAVA function named “displayfromend()” which displays the contents of the linked list from the end in a Doubly Linkedlist | | 1 | | | | |
| 34 | 7 | In a circular linked list | B | 1 | Components are all linked together in some non sequential manner. | There is no beginning and no end. | Components are arranged hierarchically. | Forward and backward traversal within the list is permitted. |
| 35 | 7 | The following is the class for which type of linked list?<br>class Node<br>{<br>Node link;<br>int value;<br>} | D | 1 | Singly Linked List | Doubly Linked List | Circular Linked List by SLL | Option A and C both |
| 36 | 7 | A variant of linked list in which last node of the list points to the first node of the list is? | C | 1 | Singly linked list | Doubly linked list | Circular linked list | Multiply linked list |
| 37 | 7 | A variation of linked list is circular linked list, in which the last node in the list points to first node of the list. One problem with this type of list is? | C | 1 | It waste memory space since the inner head already points to the first node and thus the last node does not need to point to the first node. | It is not possible to add a node at the end of the list. | It is difficult to traverse the list as the pointer of the last node is now not NULL | None of these |
| 38 | 7 | A variant of the linked list in which none of the node contains NULL pointer is? | A | 1 | Circular linked list | Singly linked list | Doubly linked list | None |
| 39 | 7 | In circular linked list, insertion of node requires modification of how many pointers while isnerting into last? | B | 1 | One Pointer | Two pointer | Three Pointer | None |
| 40 | 7 | What differentiates a circular linked list from a normal linked list? | A | 1 | You may not have the ‘next’ pointer point to null in a circular linked list | It is faster to traverse the circular linked list | You may have the ‘next’ pointer point to null in a circular linked list | Head node is known in circular linked list but not in Singly LL |
| 41 | 7 | Consider Circular Linked list With 4 nodes: 100, 200, 300, 400 and first is a pointer that pointing to node holding value 100 in the list. What should be added in place of _____ in following code to display this data?<br>void display()<br>{<br>node temp;<br>temp = first;<br>while(temp.link != first)<br>{<br>System.out.println(temp.info);<br>_____<br>}<br>System.out.println(temp.info);<br>} | A | 1 | temp = temp.link; | temp = temp.link.link; | temp = temp.info.info; | temp = temp.link.info; |
| 42 | 7 | Write a Program for inserting and deleting an element into circular linked list. | | 5 | | | | |
| 43 | 7 | Write a Program for all ( insert, delete, display) the operations in a circular linked list. | | 5 | | | | |
| 44 | 7 | Write a program to insert a node in a Circular Link List at the FIRST position. | | 5 | | | | |
| 45 | 7 | Write a function to print all odd positioned nodes from a circular linked list. (Consider 1-based indexing). | | 2 | | | | |

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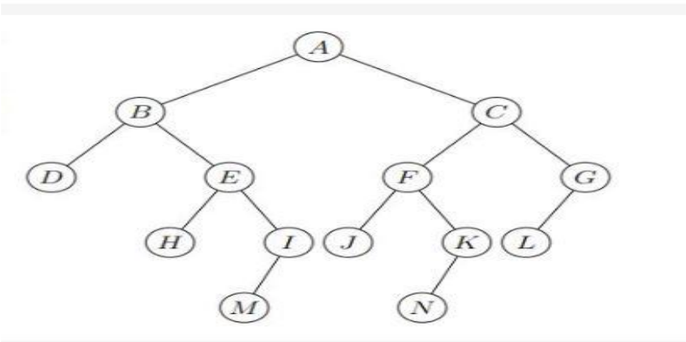
|---|---|---|---|---|---|---|---|---|
| 46 | 7 | Write JAVA functions to implement INSERT_FIRST (to insert a node at the first position), DELETE_FIRST (to delete a node from the first position), DELETE_LAST (delete a node from the last position) and TRAVERSE (to display the data in nodes) operations in circular linked list. | | 5 | | | | |
| 47 | 7 | Write a program to delete all odd positioned nodes from a circular linked list. (Consider 1-based indexing). | | 5 | | | | |
| 48 | 7 | Write a Java function that add two circular linked list and create a new linked list from it.<br>The addition will be done as following example<br>L1: 2 – 4 – 6 – 7 – 9<br>L2 : 5 – 3 – 12 – 11 – 45<br>----------<br>L3 : (2 + 3 = 5 , 4 + 12 = 16, 6 + 11 = 17, 7 + 45 = 52, 9 + 5 = 14 )<br>All are circular linked list | | 3 | | | | |
| 49 | 7 | Write a java function to delete a node at kth index from the circular linkedlist. (Assume index starts from 1)<br>It must display proper messages if the given index is not found. | | 2 | | | | |
| 50 | 7 | Which of these is an application of linked lists? | D | 1 | To implement file systems | For separate chaining in hash-tables | To implement non-binary trees | All of the mentioned |
| 51 | 7 | Linked lists are not suitable to for the implementation of? | C | 1 | Stack | Queue | Binary search | Stack and Queue |
| 52 | 8 | A tree with no nodes is known as _____ | D | 1 | General tree | NULL tree | Empty Tree | Both NULL and Empty Tree |
| 53 | 8 | Out degree of leaf node is_____. | B | 1 | 2 | 0 | 3 | 1 |
| 54 | 8 | ++a*bc*+defg is an? | D | 1 | postfix expression | infix expression | prefix expression | invalid expression |
| 55 | 8 | Assuming that the tree's height starts from 0. Then number of nodes in a tree of height h will be? | C | 1 | (2^h ) | (2^h – 1) | (2^(h+1) – 1) | (2^(h-1) – 1) |
| 56 | 8 | Suppose that the figure is a binary tree. The letters indicate the names of the nodes, not the values that are stored. What is the predecessor node, in terms of value, of the root node A?<br>  <br> | C | 1 | D | H | I | M |

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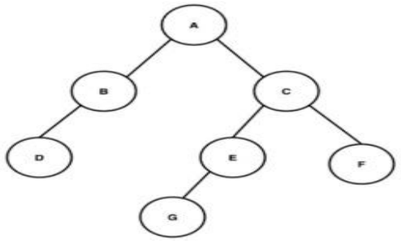
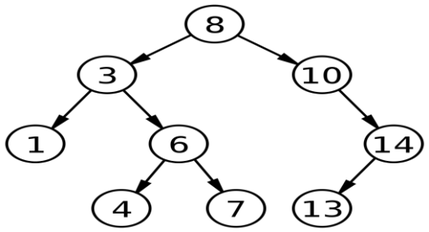
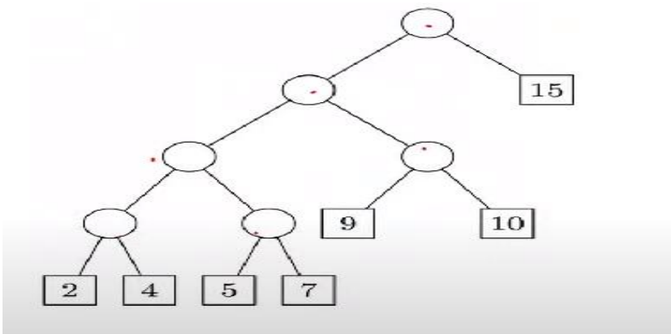
|---|---|---|---|---|---|---|---|---|
| 57 | 8 | Height of the following Tree is _____.<br>  <br> | A | 1 | 3 | 2 | 1 | 0 |
| 58 | 8 | Height of tree is _____<br>  <br> | A | 1 | 3 | 2 | 1 | 0 |
| 59 | 8 | The height of the binary tree in figure is_____.<br>  <br> | A | 1 | 4 | 5 | 6 | 7 |
| 60 | 8 | How many nodes are available at level L in complete binary tree ? | B | 1 | L^2 | 2^L | 2L^2 | 2L^2 - 1 |

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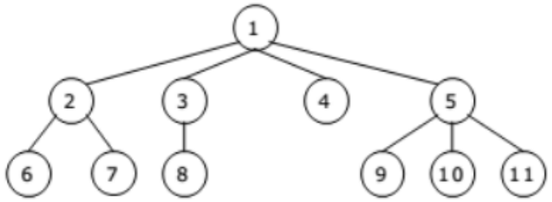
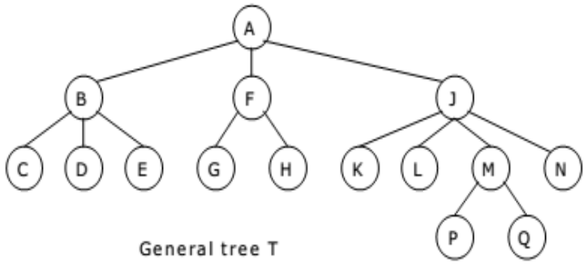
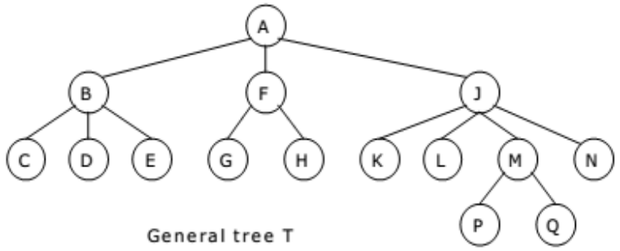
|---|---|---|---|---|---|---|---|---|
| 61 | 8 | Construct a Binary tree from given General tree. Give its inorder and preorder.   | | 3 | | | | |
| 62 | 8 | Convert given General tree into binary tree :   | | 2 | | | | |
| 63 | 8 | In a complete k-ary tree, every internal node has exactly k children or no child. The number of leaves in such a tree with 12 internal nodes is: | A | 0.5 | 12(k-1) + 1 | 11K+1 | 12k+1 | 13(k+1) |
| 64 | 8 | A strictly binary tree with 22 leaves have _____ | D | 0.5 | exactly 22 nodes | exactly 44 nodes | exactly 40 nodes | exactly 43 nodes |
| 65 | 8 | The number of structurally different possible binary trees with 8 nodes is | D | 0.5 | 1330 | 2020 | 1220 | 1430 |
| 66 | 8 | A complete n-ary tree is one in which every node has 0 or n sons. If x is the number of internal nodes of a complete n-ary tree, the number of leaves in it is given by | A | 1 | x(n-1) +1 | xn – 1 | xn + 1 | x(n+1) + 1 |
| 67 | 8 | The inorder and preorder traversal of a binary tree are d b e a f c g and a b d e c f g, respectively. The postorder traversal of the binary tree is: | A | 1 | d e b f g c a | e d b g f c a | a e b f g c d | d e f b g c a |
| 68 | 8 | What is the difference between height of node (J) and node (A)<br>Height(J) – height(A) = _____   | D | 1 | 1 | 2 | 3 | -1 |

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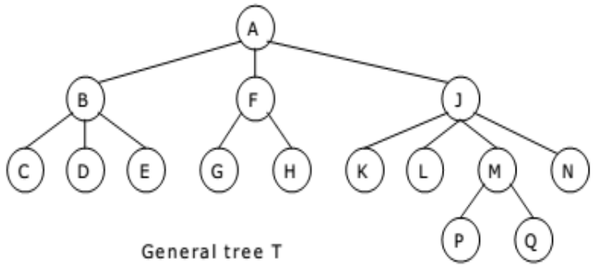
|---|---|---|---|---|---|---|---|---|
| 69 | 8 | For the general tree shown below:<br>1. Find the corresponding binary tree T.<br>2. Find the preorder, postorder, inorder traversal of T.<br>  <br><br>General tree T | | 3 | | | | |
| 70 | 8 | A strictly(Full) binary tree with 10 leaves | B | 1 | cannot have more than 19 nodes | has exactly 19 nodes | has exactly 17 nodes | has exactly 20 nodes |
| 71 | 8 | What is a full binary tree? | A | 1 | Each node has exactly zero or two children | Each node has exactly two children | All the leaves are at the same level | Each node has exactly one or two children |
| 72 | 8 | What is the maximum number of children that a binary tree node can have? | C | 1 | 0 | 1 | 2 | 3 |
| 73 | 8 | Which of the following is a true about Binary Trees | D | 1 | Every binary tree is either complete or full. | Every complete binary tree is also a full binary tree. | Every full binary tree is also a complete binary tree. | None of the above |
| 74 | 8 | In a perfect binary tree, if the height of tree is 5 then total number of nodes in tree is ___? | C | 1 | 32 | 16 | 63 | 14 |
| 75 | 8 | The preorder traversal = {7, 1, 0, 3, 2, 5, 4, 6, 9, 8, 10 },<br>Inorder traversal = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10} What is the Post order traversal? | D | 1 | 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 | 7, 6, 5, 4, 3, 2, 1, 9, 8, 10, 0 | 7, 1, 0, 3, 2, 5, 4, 6, 9, 8, 10 | 0, 2, 4, 6, 5, 3, 1, 8, 10, 9, 7 |
| 76 | 8 | In a full binary tree if number of internal nodes is I, then number of nodes N are? | A | 1 | N = 2*I + 1 | N = I + 1 | N = I – 1 | N = 2*I - 1 |
| 77 | 8 | If binary trees are represented in arrays, what formula can be used to locate parent node? | B | 1 | (i+1)/2 | (i-1)/2 | i /2 | 2*i/2 |
| 78 | 8 | Consider a binary tree T which has X number of nodes with 1 child and Y number of nodes with 2 children. If total number of nodes in tree T is N then value of N can be expressed as: | B | 1 | X+Y | X+2Y+1 | 2X+Y+1 | 2X+2Y+1 |
| 79 | 8 | In a binary tree, the number of internal nodes of degree 1 is 5, and the number of internal nodes of degree 2 is 10. The number of leaf node in the binary tree is | D | 1 | 10 | 12 | 13 | 11 |
| 80 | 8 | The no of external nodes in a full binary tree with n internal nodes is? | B | 1 | n | n+1 | n/2 | 2n |
| 81 | 8 | In a binary tree with n nodes, every node has an odd number of descendants. Every node is considered to be its own descendant. What is the number of nodes in the tree that have exactly one child? | B | 1 | 1 | 0 | (n-1)/2 | 3 |
| 82 | 8 | The postorder traversal of a binary tree is 8,9,6,7,4,5,2,3,1. The inorder of the same is 8,6,9,4,7,2,5,1,3. The height of the tree is the length of the longest path from the root to any leaf. The height of the binary tree above is_____. | A | 1 | 4 | 1 | 6 | 3 |
| 83 | 8 | What is the number of binary tree with 3 nodes which when traversed in post-order give the sequence A, B, C ? | D | 1 | 2 | 3 | 4 | 5 |
| 84 | 8 | The height of a tree is the length of the longest root-to-leaf path in it. The maximum and minimum number of nodes in a binary tree of height 4 are | D | 1 | 63,6 | 63,5 | 32,6 | 31,5 |

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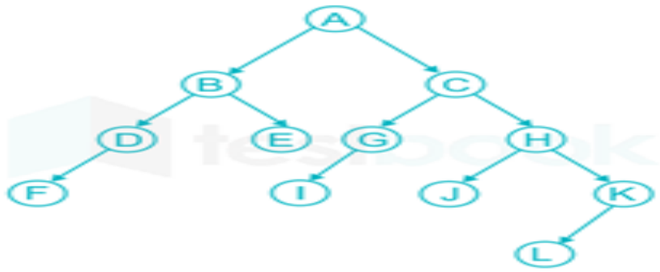
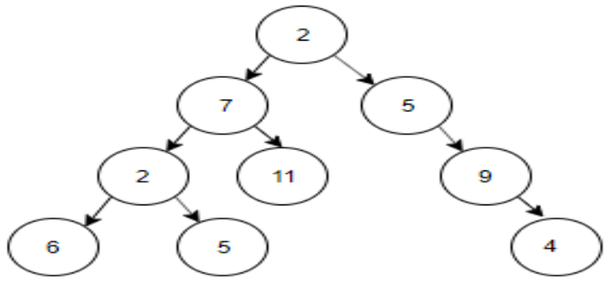
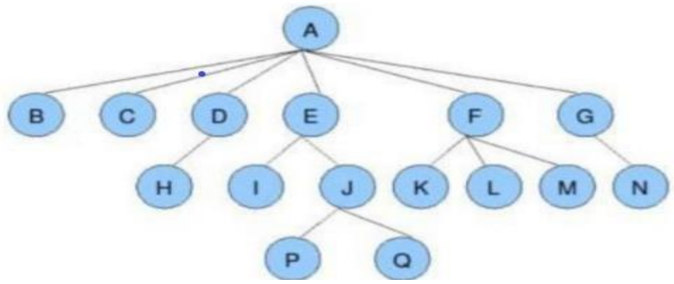
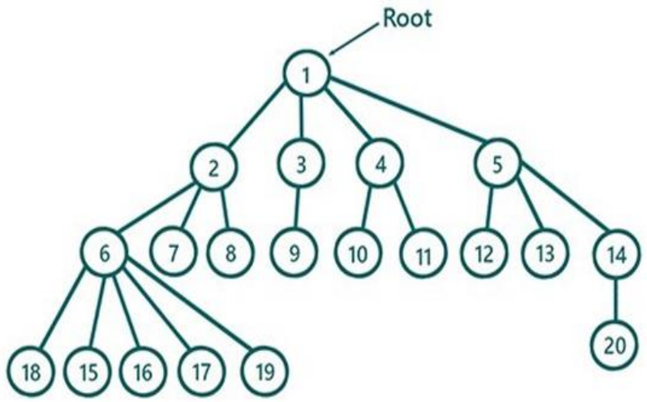
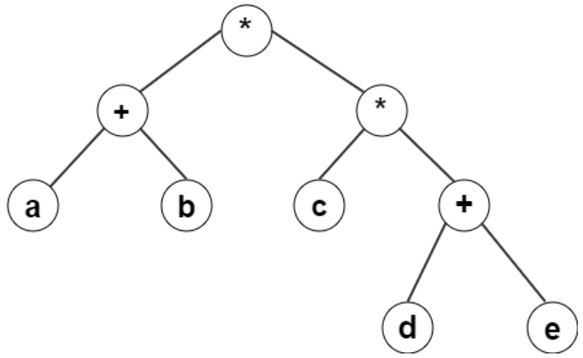
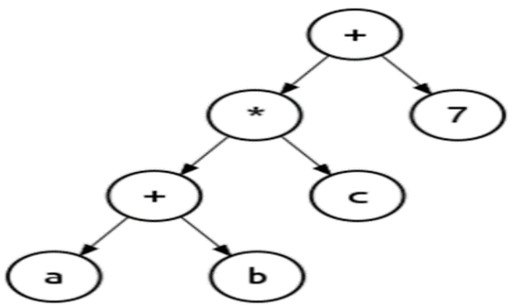
| 85 | 8 | In a binary tree, the number of internal nodes of degree 1 is 3, and the number of internal nodes of degree 2 is 6. The number of leaf node in the binary tree is | C | 0.5 | 5 | 6 | 7 | 8 |
| 86 | 8 | Question: In a binary tree, the preorder traversal is 10, 5, 2, 7, 15, 12, 20. The inorder traversal is 2, 5, 7, 10, 12, 15, 20. The height of the binary tree is 2. How many leaf nodes does the tree have? | A | 0.5 | 4 | 5 | 2 | 3 |
| 87 | 8 | In a complete k-ary tree, every internal node has exactly k children or no child. The number of leaves in such a tree with n internal nodes is: | C | 0.5 | nk | (n – 1) k+ 1 | n( k – 1) + 1 | n(k – 1) |
| 88 | 8 | A strictly binary tree with 12 leaves ……. | C | 1 | cannot have more than 19 nodes | has exactly 19 nodes | has exactly 23 nodes | has exactly 17 nodes |
| 89 | 8 | The number of structurally different possible binary trees with 6 nodes is | A | 1 | 132 | 127 | 130 | 200 |
| 90 | 8 | Which indicates pre-order traversal? | C | 1 | Left sub-tree, Right sub-tree and root | Right sub-tree, Left sub-tree and root | Root, Left sub-tree, Right sub-tree | Right sub-tree, root, Left sub-tree |
| 91 | 8 | The Preorder traversal of a tree given below is:   | A | 1 | A B D F E C G I H J K L | A B C D E G H F I J K L | A B E D F C G H I J K L | A B D F E C G I J H K L |
| 92 | 8 | Consider the following nested representation of binary trees: (X Y Z) indicates Y and Z are the left and right sub stress, respectively, of node X. Note that Y and Z may be NULL, or further nested. Which of the following represents a valid binary tree? | C | 1 | (1 2 (4 5 6 7)) | (1 (2 3 4) 5 6) 7) | (1 (2 3 4)(5 6 7)) | (1 (2 3 NULL 4) (4)) |
| 93 | 8 | Consider the following nested representation of Binary Trees. (A B C) indicates B and C are left and right subtrees of node A respectively. Note that B and C may be null or further nested.<br><br>Which of the following representation is not a valid Binary tree? | C | 1 | (1 (2 3 4) (5 6 7)) | (1 (1 (1 1 1) (1 1 1)) (1 1 1)) | (1 (2 3) 4) | (1 (2 null 4) (3 5 (6 7 null))) |

| Sr No | Unit Number | Question_Text | MCQ Answer | Marks | Option A | Option B | Option C | Option D |
|---|---|---|---|---|---|---|---|---|
| 94 | 8 | class Tree {<br>int f(TreeNode root) {<br>if (root == null \|\| (root.left == null && root.right == null))<br>return Integer.MIN_VALUE;<br><br>int l = f(root.left);<br>int r = f(root.right);<br><br>if (root.left != null \|\| root.right != null)<br>return Math.max(root.val, Math.max(l, r));<br><br>return Math.max(l, r);<br>}<br>}<br>What does the above function do? | C | 1 | Find the maximum value of all nodes in the tree | Find the maximum value of all leaf nodes | Find the maximum value of all internal nodes | Find the minimum value of all internal nodes |
| 95 | 8 | A scheme for storing binary trees in an array X is as follows. Indexing of X starts at 1 instead of 0. the root is stored at X[1]. For a node stored at X[i], the left child, if any, is stored in X[2i] and the right child, if any, in X[2i+1]. To be able to store any binary tree on n vertices the minimum size of X should be. | D | 1 | log2n | n | 2n + 1 | 2^n — 1 |
| 96 | 8 | The postorder traversal of a binary tree is 8, 9, 6, 7, 4, 5, 2, 3, 1. The inorder traversal of the same tree is 8, 6, 9, 4, 7, 2, 5, 1, 3. The height of a tree is the length of the longest path from the root to any leaf. The height of the binary tree above is _____ . | B | 1 | 3 | 4 | 5 | 6 |
| 97 | 8 | The height of a tree is the length of the longest root-to-leaf path in it. The maximum and minimum number of nodes in a binary tree of height 5 are | A | 1 | 63 and 6, respectively | 64 and 5, respectively | 32 and 6, respectively | 31 and 5, respectively |
| 98 | 8 | The height of a tree is the length of the longest root-to-leaf path in it. The maximum and minimum number of nodes in a binary tree of height 8 are | A | 1 | 511 and 9 | 512 and 8 | 255 and 9 | 256 and 9 |
| 99 | 8 | For the tree below, write the in-order traversal.<br>  <br> | | 2 | | | | |

| Sr No | Unit Number | Question_Text | MCQ Answer | Marks | Option A | Option B | Option C | Option D |
|---|---|---|---|---|---|---|---|---|
| 100 | 8 | Consider the following generic tree T.<br>  <br><br>1) Construct Binary Tree B for the given generic tree T<br>2) Find Postorder traversal of the Binary tree obtained above<br>3) Write the maximum possible nodes in the binary tree having the same height as B | | 2 | | | | |
| 101 | 8 | Draw a binary expression tree for the following:<br>a*( b + c ) + ((d + e*f)*g) | | 2 | | | | |
| 102 | 8 | The inorder and preorder traversal of binary tree are<br>d b e a f c g<br>a b d e c f g respectively. Construct a binary tree and find its postorder traversal. | | 5 | | | | |
| 103 | 8 | Construct binary tree from given postorder inorder pair.<br>After constructing Binary tree<br>(i) Find Pre-order of Tree<br>(ii) Height of node G, depth of node H<br>(iii) Which element is present at lowest level<br>Post order: - I-D-B-G-C-H-F-E-A<br>Inorder:- B-I-D-A-C-G-E-H-F<br>(Show all steps) | | 3 | | | | |
| 104 | 8 | Construct a Binary tree from the traversals given below:<br>Inorder: 1,10,11,12,13,14,15,17,18,21<br>Postorder: 1,11,12,10,14,18,21,17,15,13 | | 2 | | | | |
| 105 | 8 | Let us consider the two given arrays as pre[] = {1, 2, 4, 8, 9, 5, 3, 6, 7} and post[] = {8, 9, 4, 5, 2, 6, 7, 3, 1}. Construct a Full binary Tree from above given traversals. | | 2 | | | | |

| Sr No | Unit Number | Question_Text | MCQ Answer | Marks | Option A | Option B | Option C | Option D |
|---|---|---|---|---|---|---|---|---|
| 106 | 8 | Convert the following Generic Tree to a Binary Tree and write the Preorder and Post order traversals for the same.   | | 2 | | | | |
| 107 | 8 | Construct a binary tree from the traversals given below:<br>Inorder: 1 3 4 6 7 8 10 13 14<br>Preorder: 8 3 1 6 4 7 10 14 13 | | 5 | | | | |
| 108 | 8 | What are null nodes filled with in a threaded binary tree? | B | 1 | right node with preorder predecessor and left node with inorder successor information | inorder predecessor for left node and inorder successor for right node information | some other values randomly | inorder successor for left node and inorder predecessor for right node information |
| 109 | 8 | What is a threaded binary tree traversal? | D | 1 | a binary tree traversal using stacks | a binary tree traversal using queues | a binary tree traversal using stacks and queues | a binary tree traversal without using stacks and queues |
| 110 | 8 | In general, the node content in a threaded binary tree is _____ | A | 1 | leftchild_pointer, left_tag, data, right_tag, rightchild_pointer | leftchild_pointer, left_tag | leftchild_pointer, left_tag, right_tag, rightchild_pointer | leftchild_pointer, left_tag, data |
| 111 | 8 | What is the purpose of threading in a threaded binary tree?<br><br>1.To enable faster searching and traversal operations.<br>2.To eliminate the need for parent pointers in each node.<br>3.To enforce a strict hierarchy among the nodes in the tree. | B | 1 | Statement 1 & 2 both | Statement 1 only | Statement 3 only | Statement 2 only |
| 112 | 8 | What is the height of the expression tree which is built from the expression :<br>((A*(A + B $ C ^D))-(F-G/H*E+F))^C | D | 1 | 3 | 4 | 5 | 6 |
| 113 | 8 | The leaves of an expression tree always contain? | B | 1 | operators | operands | null | expression |

| Sr No | Unit Number | Question_Text | MCQ Answer | Marks | Option A | Option B | Option C | Option D |
|---|---|---|---|---|---|---|---|---|
| 114 | 8 | What is the postfix expression for the following expression tree?   | B | 1 | abcde++**⍰ | ab+cde+**⍰ | abc+de+**⍰ | abcd+*e+* |
| 115 | 8 | Draw a binary expression tree for the following and perform preorder traversal for the same:<br>(A + B $ C) + (D + E * F) | | 2 | | | | |
| 116 | 8 | Construct expression binary tree for: A*(A/B-B)*C+(C^D+E)*((E/A)$D)<br>Show all steps. | | 2 | | | | |
| 117 | 8 | Draw binary expression tree for given below expression and find pre-order and post order for the same (show all steps)<br>(( a * b ^ c ) – ( d + e ) ) * (f ^ k – h ) | | 3 | | | | |
| 118 | 8 | Construct an Expression tree for a*b-c*d-e/f+g-h/i and give its postorder traversal. | | 2 | | | | |
| 119 | 8 | Draw a binary expression tree for the following and perform preorder traversal for the same:<br>((A*B^C)-(D+E)*(F$K-H)) | | 2 | | | | |
| 120 | 8 | Draw a Binary expression tree for the following and perform preorder traversal: a * ( b + c ) + ( d * e ) / f + g * h | | 5 | | | | |
| 121 | 8 | Obtain the expressin tree from the following postfix representation.<br>ab + cde + ** | | 3 | | | | |
| 122 | 8 | Write the inorder, postorder and preorder traversal of the Expression Tree   | | 3 | | | | |
| 123 | 8 | Binary expression tree: a*b/c+e/f*g+k-x*y and perform the preorder traversal. | | 2 | | | | |
| 124 | 8 | Consider the expression v1*v2-(v3+v4^v5). Show the tree corresponding to the expression. | | 5 | | | | |
| 125 | 8 | Descending priority queue can be implemented using _____ | A | 1 | Max Heap | Min Heap | Max - Min Heap | General Tree |

| Sr No | Unit Number | Question_Text | MCQ Answer | Marks | Option A | Option B | Option C | Option D |
|---|---|---|---|---|---|---|---|---|
| 126 | 8 | What will be the position of 70, when a max heap is constructed on the input elements 5, 70, 45, 7, 12, 15, 13, 65, 30, 25? | A | 1 | 70 will be at root | 70 will be at last level | 70 will be at second level | 70 can be anywhere in heap |
| 127 | 8 | Heaps are primarily used for efficient extraction of the _____ | B | 1 | Minimum Element | Minimum or Maximum Element | Maximum Element | Balanced Element |
| 128 | 8 | What is the height of a complete binary tree or heap with n nodes? | A | 1 | log(n) | log(3n) | log(2n) | n |
| 129 | 8 | Which one of the following array elements represents a binary min heap? | B | 1 | 12 10 8 25 14 17 | 8 10 12 25 14 17 | 25 17 14 12 10 8 | 14 17 25 10 12 8 |
| 130 | 8 | What will be the position of 5, when a max heap is constructed on the input elements 5, 70, 45, 7, 12, 15, 13, 65, 30, 25? | B | 1 | 5 will be at root | 5 will be at last level | 5 will be at second level | 5 can be anywhere in heap |
| 131 | 8 | In an array-based implementation of a Heap, the right-child of the right-child of the node at index i, if it exists, can be found at what array location? Index of the array starts from 1. | A | 1 | 4i+3 | 2i+1 | 4i+1 | 2i+2 |
| 132 | 8 | Consider an array with the following elements<br>1 3 2 7 6 10 5 4<br>The minimum number of interchanges required to make it a min-heap is _____ | A | 1 | 1 | 2 | 3 | 4 |
| 133 | 8 | Construct a max heap for the given array of elements-<br>1, 5, 6, 8, 12, 14, 16 | | 1 | | | | |
| 134 | 8 | Consider the following max heap-<br>50, 30, 20, 15, 10, 8, 16<br>Insert a new node with value 60. | | 1 | | | | |
| 135 | 8 | Create the max heap for the following data: 1, 5, 6, 8, 12, 14, 16<br>After creating max heap kindly Perform two deletion operation. (Delete root node 2 times)<br>And find the in-order of the resultant heap. [Do not need to show all steps] | | 3 | | | | |
| 136 | 8 | 1) Construct Min heap for the following sequence of elements.<br>200,207,77,78,1,7,5,3,4,2,99,75,27,61,178,100,102<br>2) How many total swaps are needed to construct the min heap for above sequence<br>3) Perform 1 deletion on the heap constructed from above sequence and show the final heap. | | 3 | | | | |